Chapter 5: Probability

Q1. Two unbiased coins are tossed simultaneously. Find the probability of getting (i) exactly one head (ii) at least one tail.
Sample space S = {HH, HT, TH, TT}; n(S) = 4 (all outcomes equally likely).
(i) Exactly one head: Outcomes = {HT, TH}; favourable = 2
∴P(exactly one head) = 2/4 = 1/2.
(ii) At least one tail: Outcomes = {HT, TH, TT}; favourable = 3
∴P(at least one tail) = 3/4.
Answer: (i) 1/2 (ii) 3/4.

Q2. Three unbiased coins are tossed once. Find the probability of getting (i) exactly two heads (ii) at most one head.
For 3 coins: n(S) = 2^3 = 8 outcomes.
Sample space: {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}.
(i) Exactly two heads: {HHT, HTH, THH}; favourable = 3
∴P(exactly two heads) = 3/8.
(ii) At most one head (0 or 1 head):
0 head: {TTT} (1 outcome), 1 head: {HTT, THT, TTH} (3 outcomes)
Total favourable = 1 + 3 = 4
∴P(at most one head) = 4/8 = 1/2.
Answer: (i) 3/8 (ii) 1/2.

Q3. A fair die is thrown once. Find the probability of getting (i) a prime number (ii) a number greater than 4.
For one die: S = {1,2,3,4,5,6}; n(S)=6.
(i) Prime numbers on die: {2,3,5}; favourable=3
∴ P(prime) = 3/6 = 1/2.
(ii) Number > 4: {5,6}; favourable=2
∴P(>4) = 2/6 = 1/3.
Answer: (i) 1/2 (ii) 1/3.

Q4. A fair die is thrown twice. Find the probability that the sum of the numbers obtained is 8.
Two throws ⇒ total outcomes = 6×6 = 36 (ordered pairs).
Sum = 8 possible pairs: (2,6), (3,5), (4,4), (5,3), (6,2) ⇒ favourable = 5
∴P(sum = 8) = 5/36.
Answer: 5/36.

Q5. A card is drawn from a well-shuffled pack of 52 cards. Find the probability that the card drawn is (i) a heart (ii) a king (iii) neither a king nor a queen.
Total cards = 52.
(i) Hearts: 13 cards ⇒ P(heart) = 13/52 = 1/4.
(ii) Kings: 4 cards ⇒ P(king) = 4/52 = 1/13.
(iii) Neither king nor queen:
Kings = 4, Queens = 4 ⇒ (king or queen) = 8 cards
Neither king nor queen = 52 − 8 = 44 cards
∴P(neither king nor queen) = 44/52 = 11/13.
Answer: (i) 1/4 (ii) 1/13 (iii) 11/13.

Q6. Two cards are drawn successively from a pack of 52 cards without replacement. Find the probability that both cards are red.
Total red cards = 26.
P(1st red) = 26/52 = 1/2.
Without replacement, red cards left = 25, total cards left = 51.
P(2nd red | 1st red) = 25/51.
∴P(both red) = (26/52)×(25/51) = (1/2)×(25/51) = 25/102.
Answer: 25/102.

Q7. A bag contains 5 red balls, 3 blue balls and 2 green balls. One ball is drawn at random. Find the probability that it is (i) blue (ii) not red.
Total balls = 5 + 3 + 2 = 10.
(i) Blue balls = 3 ⇒ P(blue) = 3/10.
(ii) Not red means blue or green ⇒ 3 + 2 = 5
∴P(not red) = 5/10 = 1/2.
Answer: (i) 3/10 (ii) 1/2.

Q8. From the digits 1, 2, 3, 4 and 5, a number is formed by selecting one digit at random. Find the probability that the number formed is (i) an even number (ii) a prime number.
Possible digits = {1,2,3,4,5}; total outcomes = 5.
(i) Even digits: {2,4} ⇒ favourable = 2
∴ P(even) = 2/5.
(ii) Prime digits among 1–5: {2,3,5} ⇒ favourable = 3
∴ P(prime) = 3/5.
Answer: (i) 2/5 (ii) 3/5.

Q9. From the digits 0, 1, 2, 3 and 4, a two-digit number is formed without repeating any digit. Find

the probability that the number formed is divisible by 2.

Digits: {0,1,2,3,4}. Two-digit number ⇒ tens digit cannot be 0.

Total two-digit numbers without repetition: Tens digit: 4 choices (1,2,3,4). Units digit: remaining 4 choices.

Total outcomes = 4×4 = 16.

Divisible by 2 ⇒ units digit must be even: {0,2,4}.

Count favourable cases:

Units=0 ⇒ tens can be 1,2,3,4 ⇒ 4 numbers.

Units=2 ⇒ tens can be 1,3,4 ⇒ 3 numbers.

Units=4 ⇒ tens can be 1,2,3 ⇒ 3 numbers.

Total favourable = 4 + 3 + 3 = 10.

∴P(divisible by 2) = 10/16 = 5/8.

Answer: 5/8.

Q10. A box contains 12 bulbs out of which 3 are defective. One bulb is chosen at random. Find the probability that the bulb selected is not defective.

Total bulbs = 12, defective = 3 ⇒ not defective = 12 − 3 = 9.

∴P(not defective) = 9/12 = 3/4.

Answer: 3/4.